

TAMER-Augmented A* - Lightning Talk Outline (5 Minutes)

Slide 1: Title Slide (15 seconds)

Visual:

- Title: "Learning to Navigate: TAMER-Augmented A*"
- Subtitle: "Teaching Robots Human Preferences Through Feedback"
- Team: Diksha Jaiswal & Rakshit Kadam
- Simple graphic: Robot + human pointing

Talking Points:

"Hi everyone, I'm [name] and today I'll show you how we taught a path planner to respect human preferences using real-time feedback."

Slide 2: The Problem (45 seconds)

Visual:

- Left: Grid with obstacles, start (green), goal (red star)
- Middle: Shortest path (blue) going through red "hazard zone"
- Right: Desired path (green) going around hazard
- Label: "A* finds shortest path, but is it the BEST path?"

Talking Points:

"Classic motion planners like A-star are great at finding the mathematically shortest path. But look at this example—the shortest path goes straight through this hazard zone, maybe it's a crowded area or rough terrain.

As humans, we'd prefer to go around it, even if it's longer. The challenge is: how do we encode these subjective, nuanced preferences without hand-crafting complex cost functions?

That's where TAMER comes in—Training an Agent via Evaluative Reinforcement."

Slide 3: Our Approach (60 seconds)

Visual:

- **Diagram showing the loop:**
 1. A* plans path
 2. Human gives feedback (+1/-1) on path segments
 3. TAMER learns reward function
 4. A* replans with learned preferences
 5. Repeat until satisfied
- **Show modified cost function:**

Standard A*: $f(n) = g(n) + h(n)$

TAMER A*: $f(n) = g(n) + h(n) - \lambda \cdot \hat{H}(p,a)$

↑
learned human reward

Talking Points:

"Our approach combines classical A-star with TAMER's human-in-the-loop learning.

First, A-star plans a baseline path. Then a human provides simple evaluative feedback—thumbs up or thumbs down on different parts of the path.

TAMER learns a reward function from this feedback, capturing what the human likes and dislikes.

We integrate this into A-star's cost function by subtracting the learned reward. This biases the search toward human-preferred paths.

We iterate this process—plan, get feedback, replan—until the human is satisfied. It's like teaching a student through correction rather than explicit rules."

Slide 4: Results - Scenario 1 (45 seconds)

Visual:

- **Side-by-side comparison:**
 - Left panel: "Baseline A*" with blue path through red hazard zone

- Right panel: "TAMER A*" with green path avoiding hazard
- **Metrics box on each:**
 - Baseline: "Path Length: 42.3, Hazard Cells: 18"
 - TAMER: "Path Length: 45.1, Hazard Cells: 0" ← highlight this
- **Small training curve below:**
 - X-axis: Iterations (0-5)
 - Y-axis: Hazard Cells
 - Shows decline from 18 to 0

Talking Points:

"Here's our first scenario—implicit hazard avoidance. The red zone isn't an obstacle, so baseline A-star doesn't care about it.

After just 5 training iterations with simulated human feedback, look at the TAMER path—it completely avoids the hazard zone!

Yes, the path is slightly longer—45 versus 42 units—but it reduces hazard exposure from 18 cells to zero. That's exactly the trade-off a human would make.

The training curve shows this happens quickly—TAMER converges in about 3-4 iterations."

Slide 5: Results Summary (45 seconds)

Visual:

- **Table showing all 3 scenarios:**

Scenario	Baseline Hazards	TAMER Hazards	Improvement
1. Hazard Avoidance	18	0	-18 (100%)
2. Safety Margin	2	0	-2 (100%)
3. Route Preference	12	1	-11 (92%)

- **Key Findings box:** ✓ 80-100% hazard reduction ✓ Only 5-15% path length increase ✓ Converges in 3-5 iterations ✓ No manual cost engineering needed

Talking Points:

"Across all three scenarios—hazard avoidance, safety margins, and subjective route preferences—TAMER

consistently learned to respect human preferences.

We saw 80 to 100 percent reduction in undesirable path segments, with only a modest 5 to 15 percent increase in path length.

The key insight is that we captured these preferences without writing any explicit cost functions. The human feedback did all the teaching.

This opens up exciting possibilities for robots that adapt to individual user preferences in real-time."

Slide 6: Conclusion & Extensions (30 seconds)

Visual:

- **"What We Showed" column:** ✓ TAMER + A* successfully learns preferences ✓ Human feedback shapes planning behavior ✓ Practical trade-offs (path length vs. preference)
- **"Future Work" column:** → Extend to RRT* and continuous spaces → Deploy on real mobile robots → Multi-objective optimization → Deep learning reward models
- **Visual:** Small image of robot in real environment

Talking Points:

"To summarize: we successfully demonstrated that combining TAMER with classical planning lets robots learn human preferences from simple feedback, no complex programming required.

Next steps include extending this to more advanced planners like RRT-star, deploying on real robots, and exploring deep learning reward models for richer feature representations.

Thank you! I'm happy to take questions."

Backup Slides (If Time Allows)

Technical Details

- Feature extraction for reward model
- Lambda parameter tuning (0.5-1.0 works well)
- 8-connected grid with diagonal moves

Interactive Demo

- Show video/GIF of click-to-feedback interface

- "In practice, a user would click on good and bad path segments"

Implementation Stats

- ~500 lines of Python
 - Runs in real-time on laptop
 - All code available on GitHub [if applicable]
-

Presentation Tips

Timing Breakdown:

- Slide 1: 15 sec (0:15 total)
- Slide 2: 45 sec (1:00 total)
- Slide 3: 60 sec (2:00 total)
- Slide 4: 45 sec (2:45 total)
- Slide 5: 45 sec (3:30 total)
- Slide 6: 30 sec (4:00 total)
- **Buffer: 1 minute for transitions/questions**

Delivery Tips:

DO:

- Speak clearly and make eye contact
- Use laser pointer to highlight hazard zones in comparisons
- Emphasize the visual difference between baseline and TAMER paths
- Show enthusiasm about the results
- Pause briefly between slides for comprehension

DON'T:

- Rush through the math equations
- Read bullet points verbatim
- Apologize for "simple" implementation

- Get bogged down in technical details
- Go over 5 minutes

Key Messages to Emphasize:

1. **Problem is relatable:** "We all prefer certain routes over others"
2. **Solution is elegant:** "Simple feedback, no complex programming"
3. **Results are convincing:** "100% hazard reduction with minimal cost"
4. **Future is exciting:** "Robots that adapt to individual preferences"

Handling Questions:

Expected Questions:

Q: "How does this scale to real environments?" A: "Great question. Our grid-based proof-of-concept demonstrates the core idea. For real robots, we'd extend to continuous spaces using RRT* or similar planners, and the TAMER reward model would use richer features like LIDAR data or camera input."

Q: "How many feedback samples do you need?" A: "In our experiments, 3-5 iterations with about 10-20 feedback signals per iteration was sufficient. That's roughly 50-100 total feedback points, which takes a human just a few minutes."

Q: "What if different humans give conflicting feedback?" A: "Excellent point. In practice, you'd train per-user to capture individual preferences, or aggregate feedback from multiple users to learn general preferences. You could also weight feedback by user expertise."

Q: "Why not just use reinforcement learning?" A: "Traditional RL requires millions of samples and a predefined reward function. TAMER's key advantage is that it learns directly from human evaluative feedback in just minutes, making it much more practical for customization."

Q: "What about safety guarantees?" A: "Important consideration. We could add hard constraints to A^* (e.g., never enter certain regions) while using TAMER for soft preferences. This separates safety-critical requirements from subjective preferences."

Q: "Could this work in 3D or for aerial vehicles?" A: "Absolutely. The framework is dimensionality-agnostic. For drones, we'd need features capturing altitude preferences, terrain clearance, etc., but the TAMER learning mechanism stays the same."

Slide Design Recommendations

Color Scheme:

- **Background:** Clean white or light gray
- **Obstacles:** Black or dark gray
- **Hazard zones:** Light red (rgba: 255, 0, 0, 0.25)
- **Baseline path:** Blue ( #2196F3)
- **TAMER path:** Green ( #4CAF50)
- **Start:** Green circle
- **Goal:** Red star

Fonts:

- **Title:** Bold, 36-40pt
- **Headers:** Bold, 28-32pt
- **Body:** Regular, 20-24pt
- **Captions:** Regular, 16-18pt
- **Font family:** Sans-serif (Arial, Helvetica, or Calibri)

Visual Hierarchy:

1. Most important: Side-by-side path comparisons
2. Supporting: Metrics boxes
3. Background: Grid and obstacles

Animation (Optional):

- Slide 3: Animate the feedback loop cycle
 - Slide 4: Fade in baseline first, then TAMER (shows contrast)
 - Slide 5: Bars grow to show hazard reduction
-

Practice Script (Read Aloud)

Full 4-Minute Script:

"Good morning everyone. Today I'm presenting our work on teaching robots to navigate according to human preferences.

[SLIDE 1]

This is Learning to Navigate: TAMER-Augmented A-star, work by myself and my teammate Rakshit Kadam.

[SLIDE 2]

So here's the problem. Classic path planners like A-star are excellent at finding the mathematically shortest path between two points. But as you can see in this example, the shortest path—shown in blue—goes straight through this red hazard zone. Maybe that's a crowded area, or rough terrain, or just somewhere we'd rather avoid.

As humans, we'd clearly prefer to go around it. But encoding these subjective preferences into a traditional cost function is really challenging. That's where TAMER comes in.

[SLIDE 3]

TAMER stands for Training an Agent via Evaluative Reinforcement, and here's how we integrated it with A-star.

First, A-star plans an initial path. Then a human provides simple feedback—thumbs up or thumbs down on different parts of the path. TAMER learns a reward function from this feedback, and we integrate it into A-star's cost function by subtracting this learned reward. This biases the search toward human-preferred regions.

We iterate—plan, feedback, replan—until the human is satisfied. It's like teaching through correction rather than writing explicit rules.

[SLIDE 4]

Here are our results for the hazard avoidance scenario. On the left is baseline A-star, happily cutting through the hazard zone. On the right, after just five training iterations, our TAMER-augmented planner completely avoids the hazard.

Yes, the path is slightly longer—45 versus 42 units—but hazard exposure drops from 18 cells to zero. That's exactly the trade-off a human would make. And as the training curve shows, this convergence happens quickly.

[SLIDE 5]

Across all three test scenarios—hazard avoidance, safety margins, and route preferences—TAMER achieved 80 to 100 percent reduction in undesirable path segments, with only 5 to 15 percent increase in path length.

The key insight: we captured complex human preferences without writing any explicit cost functions. The human feedback did all the teaching.

[SLIDE 6]

To conclude, we successfully demonstrated that combining TAMER with classical planning enables robots to learn human preferences from simple feedback.

Future work includes extending to continuous spaces, deploying on real robots, and exploring deep learning for richer reward models.

Thank you, and I'm happy to take questions."

[END - 4 minutes]

Post-Presentation Tasks

After the Talk:

1. ✓ Save your slides as PDF
2. ✓ Upload code to GitHub (if allowed)
3. ✓ Share visualizations with classmates
4. ✓ Document feedback for final report
5. ✓ Note questions asked for paper discussion section

For Final Report:

- Expand on methodology details
- Add more quantitative analysis
- Include ablation studies (different λ values)
- Compare with other reward shaping methods
- Discuss limitations more thoroughly

Confidence Boosters

You've Got This Because:

1. ✓ **Working implementation** - Your code runs and produces results

2. ✓ **Clear visuals** - The comparisons speak for themselves
3. ✓ **Solid motivation** - Everyone understands preferring one route over another
4. ✓ **Concrete results** - 100% hazard reduction is impressive
5. ✓ **Good story arc** - Problem → Approach → Results → Future

If You Get Nervous:

- Take a deep breath before starting
- Focus on one friendly face in the audience
- Remember: You know this material better than anyone in the room
- It's okay to say "I don't know, but that's a great question for future work"
- Smile—you worked hard and should be proud!

Emergency Backup:

If technical difficulties occur:

- Have slides on USB drive
 - Have PDF version ready
 - Have slides on laptop AND uploaded to cloud
 - Practice presenting without animations/videos
-

Final Checklist (Night Before)

Technical:

- ☐ All PNG files generated and look correct
- ☐ Slides exported to PDF
- ☐ Laptop fully charged
- ☐ USB backup created
- ☐ Presentation tested on presentation computer (if possible)

Content:

- ☐ Practiced timing (aim for 4:00-4:30 to be safe)
- ☐ Can explain each equation
- ☐ Know all numbers in metrics

- ☐ Prepared for top 3 expected questions

Logistics:

- ☐ Know room location
 - ☐ Arrive 10 minutes early
 - ☐ Have water bottle
 - ☐ Dress professionally
 - ☐ Turn off phone
-

Motivation

Remember: This project is **genuinely cool**. You're showing how robots can learn human preferences through natural interaction, without complex programming. That's the future of human-robot collaboration.

You've implemented something impressive in just 10 days. Be confident, be clear, and enjoy sharing your work!

You've got this! 🚀